

Jump Planning and Airborne Attitude Control of Bipedal Wheel-Legged Robots

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Abstract—This letter investigates the jumping mechanism of a bipedal wheel-legged robot, proposing a comprehensive trajectory planning and control system that successfully achieves jumps of varying heights while significantly improving stability and reliability. Given the underactuated, non-linear and transient dynamics inherent in the jumping process, precisely controlling the jump height while ensuring stability is a challenging task. To address this, a trajectory planning method is introduced to achieve target jump heights, complemented by an impedance controller for trajectory tracking and a jump state observer for accurate stage identification. To mitigate the issue of aerial attitude divergence, a novel airborne dynamics model is proposed to characterize the robot's fundamental aerial dynamics, enabling effective airborne attitude control and ensuring high stability during the jump. Both simulations and prototype experiments validate the effectiveness of the proposed method.

Index Terms—Legged robots, wheeled robots, dynamics.

I. INTRODUCTION

WHEEL-legged robots, which combine the mobility of wheeled robots with the dexterity of legged systems [1], have witnessed rapid advancements in recent years. Representative platforms include the Ascento series developed by ETH Zurich [2], [3], [4], as well as WLR-3P [5], SR600-II [6], DIABLO [7], and Ollie [8]. These robots, featuring innovative bipedal wheel-legged architectures, demonstrate outstanding performance in dynamic balance and adaptation to complex terrains. With continued research progress, the development of dynamic models such as the wheeled spring-loaded inverted pendulum (W-SLIP) [9], [10], the wheeled linear inverted pendulum (W-LIP) [8], [11], and full-body dynamics models [12], [13], along with the implementation of advanced control strategies including LQR [7], MPC [14], and adaptive LQR [13], has significantly driven both theoretical breakthroughs and engineering

applications in the dynamic balance control of wheel-legged robots.

Building upon the foundation of stable balance control, recent efforts have sought to enhance the obstacle-crossing capabilities of wheel-legged bipedal robots in unstructured environments [15]. Although studies on jumping control remain relatively limited, notable progress has been reported. The Ascento robot [2] implemented a sequential jumping controller capable of generating jumping motions; however, it lacked effective stability guarantees and comprehensive planning, resulting in poor control over jump height. Lu et al. [15] proposed a jump trajectory planning method with soft-landing capability, utilizing virtual model control and virtual force compensation for trajectory tracking. Nevertheless, this approach overlooked body stability and exhibited noticeable discrepancies between planned and actual trajectories. Che et al. [9] introduced the W-SLIP model and a hierarchical planning and control framework for wheel-legged bipedal jumping, which demonstrated promising simulation results but has yet to be validated on hardware. Similarly, Zhuang et al. [16] developed a novel wheel-legged jumping dynamics model combined with Bayesian torque optimization, but this method remains confined to simulation due to computational challenges. Checao et al. [17] designed a hierarchical controller based on full-body dynamics for uneven terrain, achieving 0.55 m jumps in simulation. However, real-world experiments revealed a significant performance gap, with the robot achieving only 0.05 m jumps and exhibiting substantial body pitch oscillations, undermining stability. Yi et al. [18] employed a PD controller with predefined leg-height sequences and spline interpolation, enabling physical jumps of approximately 6 cm, but without precise height control. Zhanhao et al. [19] implemented a segmented control strategy for standing jumps, yet the reliance on body rotation for in-flight attitude adjustment led to excessive yaw deviations, severely compromising stability.

In summary, various methods have been proposed and preliminarily verified in the field of bipedal wheel-legged robot jump control. However, most of the research is still at the theoretical analysis and simulation stage and lacks sufficient real-machine experimental data. While some existing methods can perform basic jumping functions, they still have issues with poor jumping stability, complex algorithm solutions, insufficient consideration of robot attitude control and difficulty reaching the expected jumping height in actual prototype tests. This seriously restricts the potential of robots in complex environments. For this reason, this paper's research focuses on achieving robot jumping control

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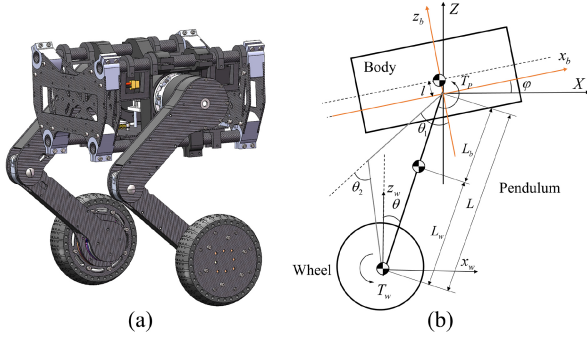


Fig. 1. (a) Physical model of the biped wheel-legged robot. (b) Simplified dynamics model of the biped wheel-legged robot.

at different heights, as well as improving jumping stability and reliability, with the main contributions being as follows:

- 1) A trajectory planning method is proposed to achieve different jump heights. A jumping state observer is developed to determine the jumping stage and an impedance controller is designed to track the entire jumping process. This successfully accomplishes the jumping control task of the robot at different desired heights.
- 2) A new airborne dynamics model is proposed to characterize the fundamental dynamics of the airborne phase of a bipedal wheel-legged robot during the jumping process. Based on this model, an airborne attitude controller has been designed to effectively regulate the robot's attitude during this phase, significantly improving the stability and reliability of the jumping process.
- 3) The jump and airborne attitude control methods are successfully validated on a physical prototype, with experimental results demonstrating the effectiveness of the proposed methods.

II. MOTION CONTROL

The bipedal wheel-legged robot designed in this study features a single leg with three degrees of freedom. Both the knee and hip joint motors are mounted above the body, resulting in low rotational inertia for the leg. In the modeling process, the freedom of the hip joint is retained, and the entire leg is simplified as a pendulum, while the body is treated as a point mass. In addition, we ignore joint friction damping and neglect motor dynamics to simplify the model. As a result, the entire robot is simplified into an inverted pendulum model, as shown in Fig. 1.

Based on the model and the derivation procedure presented in [7], [20], the state space form of the linearized dynamics equations of the robot under equilibrium conditions can be obtained. The controller is selected as LQR and the optimal control input is determined by minimizing the quadratic cost function of the linear system. The cost function and the control input can be expressed as:

$$\begin{cases} \min(J) = \int_0^\infty (x^T Q x + u^T R u) dt \\ u = -K(x - x_{ref}) \end{cases} \quad (1)$$

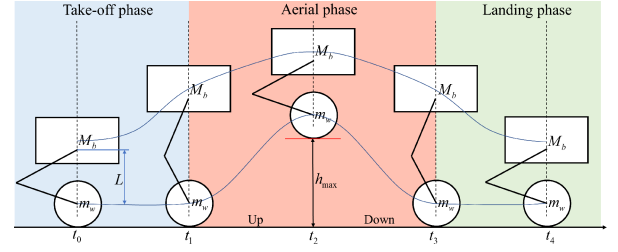


Fig. 2. Schematic diagram of jumping process and trajectory planning.

where $Q \in \mathbb{R}^{6 \times 6}$ is the state weight matrix, $R \in \mathbb{R}^{2 \times 2}$ is the control weight matrix, $u \in \mathbb{R}^{2 \times 1}$ is the hip and drive wheel torque, x_{ref} is the reference trajectory. K is the control gain, which can be obtained by solving the Riccati equation.

III. JUMP PLANNING AND CONTROL

A. Jump Planning

The main focus of this work is the planning and control of robot jumping. To achieve a specified jump height, we model and analyze the jumping process of the robot. Given the stability of the bipedal wheel-legged robot's balance control, we focus solely on vertical trajectory planning for the robot's height. As shown in Fig. 2, we divide the entire process into three main phases: the takeoff phase, the aerial phase, and the landing phase. The time intervals are defined as follows: $t_0 - t_1$ represents the acceleration phase during takeoff, $t_1 - t_2$ is the ascent phase during the aerial phase, $t_2 - t_3$ is the descent phase, and $t_3 - t_4$ is the landing buffer phase.

At the moment t_1 , we assume that the robot leg stiffness is infinite, and the driving wheel moves upward driven by the body and the leg, at which time there is:

$$\frac{1}{2} M_b \dot{z}_{t_1-}^2 = \frac{1}{2} M \dot{z}_{t_1+}^2 \quad (2)$$

where $\dot{z}_{t_1-}$ is the vertical upward velocity of the body of the robot in the instant before the moment of t_1 ; $\dot{z}_{t_1+}$ is the vertical upward velocity of the robot in the instant after the moment of t_1 ; M_b is the sum of the original body and knee joint mass, M is the mass of the whole machine.

In the time period $t_1 - t_2$, the robot has only the motion upward under inertia and the motion of leg contraction for the motion under under inertia:

$$\begin{cases} \dot{z}_{t_1+}^2 = 2gh_1 \\ \dot{z}_{t_1+} = g(t_2 - t_1) \end{cases} \quad (3)$$

Neglecting the leg mass and analyzing the single motion process of the robot's leg contraction in the air on the effect of height alone, as shown in Fig. 3, there is:

$$\begin{cases} m_w \Delta h = M_b (\Delta h_2 - \Delta h) \\ \Delta h_2 = l_1 (\cos(\alpha_{t1}) - \cos(\alpha_{t2})) \\ \quad + l_2 (\cos(\alpha_{t2} + \beta_{t2}) - \cos(\alpha_{t1} + \beta_{t1})) \end{cases} \quad (4)$$

where m_w represents the mass of the driving wheel. Δh_2 enotes the vertical displacement caused by the contraction of the robot's leg. Δh corresponds to the displacement of the driving wheel

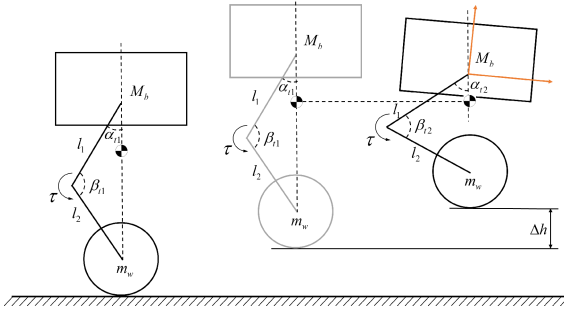


Fig. 3. Modeling of the leg contraction process during the airborne phase.

resulting from leg contraction. l_1 and l_2 represent the lengths of the thigh and shank, respectively. $\alpha_{t1}, \alpha_{t2}, \beta_{t1}, \beta_{t2}$ are the angles between the thigh and the vertical direction at moments t_1 and t_2 , respectively, and the angle between the thigh and the calf.

Then the maximum distance from the ground that the robot can reach during the jumping process:

$$h_{\max} = h_1 + \Delta h \quad (5)$$

From (2)–(5), the relationship between the maximum height of the robot's jump and the velocity of the robot's body at the instant before it leaves the ground is:

$$h_{\max} = \frac{M_b \dot{z}_{t1}^2}{2Mg} + \frac{M_b \Delta h_2}{M} \quad (6)$$

1) *Take-off phase*: According to (6), we can realize the maximum jump height by controlling the velocity of the robot body before it leaves the ground. Here, we plan the trajectory of the robot body for the purpose of planning and, to ensure the smoothness of the motion velocity and acceleration, a cubic polynomial is used for planning:

$$\begin{cases} z(t) = a_1(t - t_0)^3 + b_1(t - t_0)^2 + c_1(t - t_0) + d_1 \\ \dot{z}(t) = 3a_1(t - t_0)^2 + 2b_1(t - t_0) + d_1 \\ \ddot{z}(t) = 6a_1(t - t_0) + 2b_1 \end{cases}, t_0 \leq t \leq t_1 \quad (7)$$

Once the trajectory is determined, the force F_1 exerted by the legs on the body can be controlled to track the vertical trajectory. Additionally, to eliminate potential tipping during the tracking process, an additional controller is required to regulate the body's roll angle θ_r .

$$F_1' = K_p(0 - \theta_r) + K_d(0 - \dot{\theta}_r) \quad (8)$$

Thus the force on the Z-axis required by the left and right side legs:

$$F_{z1L} = \frac{F_1}{2} + F_1', F_{z1R} = \frac{F_1}{2} - F_1' \quad (9)$$

2) *Aerial phase*: The robot primarily follows its inertial motion from before takeoff. It is necessary to control both legs to move according to the predefined trajectory. Furthermore, efforts should be made to ensure that when the body reaches its highest

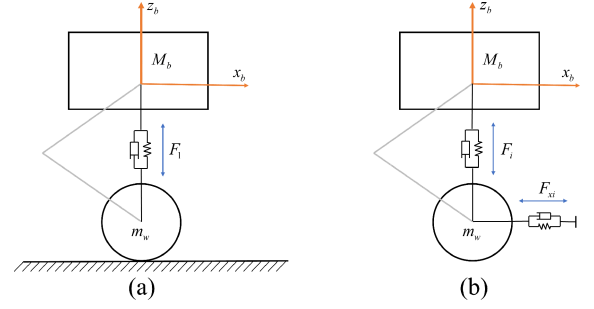


Fig. 4. (a) Leg impedance controller model for the take-off phase. (b) Leg impedance controller model for the aerial phase.

point, the legs are also contracted to their shortest length. A cubic polynomial is still used for trajectory planning.

$$\begin{cases} z(t) = a_i(t - t_i)^3 + b_i(t - t_i)^2 + c_i(t - t_i) + d_i \\ \dot{z}(t) = 3a_i(t - t_i)^2 + 2b_i(t - t_i) + d_i \\ \ddot{z}(t) = 6a_i(t - t_i) + 2b_i \end{cases}, t_1 \leq t \leq t_3, i = 2, 3 \quad (10)$$

At this point, the forces on the Z-axis required by the left and right legs are F_{z2L} and F_{z2R} in the airborne ascent phase and F_{z3L} and F_{z3R} in the airborne descent phase.

3) *Landing phase*: When the robot detects landing, the buffering controller must be activated immediately. The legs and wheels are used to absorb the initial impact and prevent oscillations. To reduce the impact, the robot can mimic the human squatting motion for a safe landing [21]. In this study, a controller with variable stiffness and damping is employed to control the robot's landing [22], focusing solely on vertical control. Thus, the virtual force exerted by the robot's legs is:

$$F_{z4L} = F_{z4R} = K_{pz}(z_d - z_f) + K_{dz}(\dot{z}_d - \dot{z}_f) \quad (11)$$

where K_{pz} and K_{dz} are the virtual stiffness and damping along the z-axis, which are assumed to be linear models in the buffer stage:

$$\begin{cases} K_{pz} = K_{pz0} + k_{ps}(t - t_3) \\ K_{dz} = K_{dz0} + k_{ds}(t - t_3) \end{cases}, t_3 \leq t \leq t_4 \quad (12)$$

where K_{pz0}, K_{dz0} are the initial values of stiffness and damping in the z-axis direction at the moment t_3 , k_{ps} and k_{ds} are the rates of change of stiffness and damping.

B. Jump Trajectory Tracker

1) *Take-off phase*: To achieve the required takeoff velocity for the target jump height, precise tracking of the robot's velocity and position in the vertical direction is necessary. By introducing the characteristics of impedance control [23], [24], the robot's body is simplified as a single-degree-of-freedom system in the vertical direction, as illustrated in Fig. 4(a). The system dynamics can be expressed by the following equation:

$$M_b \ddot{z} = F - F_{ext} - M_b g \quad (13)$$

where F represents the output force calculated by the controller and F_{ext} is the contact force between the robot and the environment.

Introducing the core formula for impedance control:

$$\begin{cases} F_{ext} = M_d \ddot{e} + D_d \dot{e} + K_d e \\ e = z_d - z \end{cases} \quad (14)$$

where M_d, D_d, K_d represent the inertia characteristics, damping characteristics and stiffness characteristics respectively, z is the actual position and z_d is the desired position. Combined with (13)–(14), the controller outputs for the jumping phase can be obtained:

$$F_{z1} = (M_d - M_b) \ddot{e} + M_b \ddot{z}_d + D_d \dot{e} + K_d e + M_b g \quad (15)$$

2) *Aerial phase*: The robot is treated as a two-degree-of-freedom control system based on a floating base, with the drive wheels at the foot being the main control target. At this point, the robot is airborne, and no controller is available to provide balance in the horizontal direction. Therefore, an additional horizontal controller is required to constrain the trajectory of the foot's drive wheel. In this phase, we ignore the body's posture and introduce the concept of virtual model control to regulate the foot's motion. As shown in Fig. 4(b), the following controller can be derived:

$$\begin{cases} F_{zi} = (M_d - m_w) \ddot{e} + m_w \ddot{z}_d + D_d \dot{e} + K_d e + m_w g \\ F_{xi} = K_{px}(x_d - x) + K_{dx}(\dot{x}_d - \dot{x}) \end{cases}, i = 2, 3 \quad (16)$$

Through the above analysis, we have obtained the required trajectories of the body and wheels and the corresponding magnitude of the control forces at different time intervals. Therefore, it is only necessary to map them back to the joints through the Jacobi matrix to accomplish the control:

$$\begin{bmatrix} \tau_1 \\ \tau_2 \end{bmatrix}_j = J^T \begin{bmatrix} F_{xi} \\ F_{zi} \end{bmatrix}_j, i = 1, 2, 3, 4, j = L, R \quad (17)$$

where τ_{1j} and τ_{2j} are the hip and knee moments on the right and left legs, J is the robotic single-leg Jacobi matrix, F_{xi} is the force required at the foot end in the horizontal X-direction, and F_{zi} is the force required at the foot end in the vertical Z-direction.

C. Jump State Observer

After completing the trajectory planning for the jumping phase, it is necessary to determine the current phase in order to follow the planned trajectory. This section presents a state observer for the bipedal wheel-legged robot's jump, which is used to identify the phase during the robot's jumping process. Since it is not feasible to install pressure sensors on the drive wheels to measure the wheel-ground force, we adopt a multi-sensor fusion approach for comprehensive determination. Initially, the ground reaction force can be computed through sensor fusion.

$$\begin{cases} F_N = F_{lw} \cos(\theta) + m_w g + m_w \ddot{z}_w \\ \ddot{z}_w = \frac{d^2}{dt^2}(z_M - L \cos(\theta)) \end{cases} \quad (18)$$

where F_N is the support force of the ground on the wheel, F_{lw} is the force of the robot leg on the wheel, $\ddot{z}_w$ is the wheel acceleration in the vertical direction.

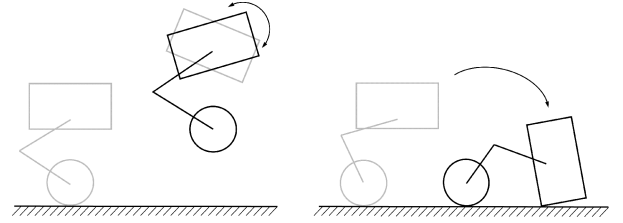


Fig. 5. Schematic representation of robot jump instability and failure.

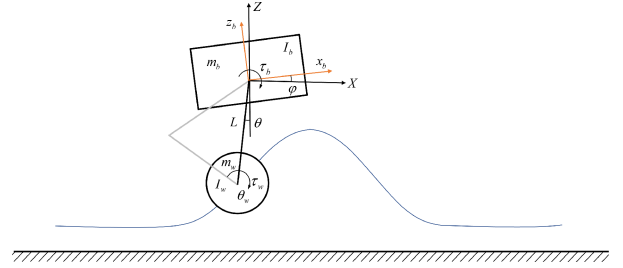


Fig. 6. Airborne dynamics model.

Additionally, during the robot's motion, it may encounter situations such as single-leg contact or unstable posture. Relying solely on changes in the wheel-ground force cannot accurately detect the robot's takeoff or touchdown states. Therefore, this study also incorporates the detection of the body's acceleration to make a comprehensive judgment regarding the robot's aerial or landing state. Thus, the vertical support force F_N exerted by the drive wheel and the body's vertical acceleration a_z are used as criteria to determine whether the robot is airborne.

$$\xi = \begin{cases} 1, (F_N > f_{nb} \ \& \ a_z > a_{gb}) \\ 0, (F_N < f_{nb} \ \& \ a_z < a_{gb}) \end{cases} \quad (19)$$

where f_{nb} is the support force threshold for determining whether the robot is off the ground, a_{gb} is the Z-axis acceleration threshold for determining whether the robot is off the ground; ξ is the result of the judgment, $\xi = 1$ means that the robot is not off the ground, $\xi = 0$ means that the robot is off the ground. In addition, during the entire jumping process, the judgment of leaving and touching the ground is only performed once.

D. Airborne Dynamics Model

During the robot's aerial phase, the system is in a state of angular momentum conservation. The body and legs experience significant oscillations due to the extension and contraction of the legs during the jump. Moreover, if the robot jumps while unbalanced, there is a high likelihood of the robot flipping in the air, which can lead to a failed jump, as shown in Fig. 5.

To address this, we propose a novel airborne dynamics model to describe the basic dynamic characteristics of the bipedal wheel-legged robot in flight. Based on this model, to control the attitude adjustment of the robot in the aerial, so as to realize the high stability of jumping. We simplify the robot's model during the aerial phase and establish the model shown in Fig. 6, which consists of three main components: the body, pendulum,

and drive wheel. This planar model has two active degrees of freedom, namely the hip joint and the wheel joint. We assume that the zero potential energy point is located at the center of rotation of the hip joint, the center of mass of the drive wheel coincides with its center of rotation, and the center of mass of the pendulum is at its geometric center. The base coordinate system is also defined at the center of rotation of the hip joint.

The system's total kinetic energy primarily consists of four components: the rotational kinetic energy of the main body, the translational and rotational kinetic energy of the pendulum rod, and the translational and rotational kinetic energy of the driving wheels.

$$T = \frac{1}{2}I_b\dot{\varphi}^2 + \left(\frac{1}{2}I_p\dot{\theta}^2 + \frac{1}{2}m_pv_p^2\right) + \left(\frac{1}{2}I_w\dot{\theta}_w^2 + \frac{1}{2}m_wv_w^2\right) \quad (20)$$

where I_w , I_p and I_b are the moment of inertia of the drive wheel, the pendulum, and the body, respectively. φ and $\dot{\varphi}$ are the body pitch angle and pitch angular velocity. θ and $\dot{\theta}$ are the pendulum angle and angular velocity. θ_w and $\dot{\theta}_w$ are the angle of rotation and angular velocity of the driving wheel. v_p and v_w are the translation velocities of the pendulum rod and the driving wheel, respectively, which can be obtained through kinematic differentiation. l represents the distance from the center of mass of the machine body to the rotation center of the machine body.

The potential energy of the system is:

$$V = m_bgl\sin(\varphi) + m_pg z_p + m_w g z_w \quad (21)$$

where the displacements of the pendulum rod and the driving wheel are:

$$\begin{cases} x_p = -L/2\sin(\theta), z_p = -L/2\cos(\theta) \\ x_w = -L\sin(\theta), z_w = -L\cos(\theta) \end{cases} \quad (22)$$

According to the Lagrange equation:

$$\begin{cases} L = T - V \\ \frac{d}{dt}\frac{\partial L}{\partial \dot{q}} - \frac{\partial L}{\partial q} = Q_i \end{cases} \quad (23)$$

Solve the final equations for the airborne dynamics model:

$$\begin{cases} \ddot{\varphi} = \frac{1}{I_b}(\tau_b - m_bgl\cos(\varphi)) \\ \ddot{\theta} = \frac{\tau_b - \tau_w - (\frac{m_p}{2} + m_w)g\cdot L\cdot\sin\theta}{(\frac{1}{4}m_p + m_w)L^2 + I_p} \\ \ddot{\theta}_w = \frac{1}{I_w}\tau_w \end{cases} \quad (24)$$

where τ_b is the hip joint output moment, τ_w is the drive wheel output moment.

E. Airborne Attitude Controller

During the aerial phase, our control objective is to quickly bring the body and legs back to a stable state, making the rotation angle of the drive wheel no longer the focus of control. As shown in (24), when the robot is airborne, the torque of the hip joint influences the body pitch angle, and the torques of the hip joint and drive wheel jointly affect the swing angle of the pendulum, resulting in a coupling relationship between the two. Despite this coupling, due to the strong robustness of PID control, it remains effective in achieving posture control. Furthermore, PID control offers a simple structure and ease of implementation. To enhance

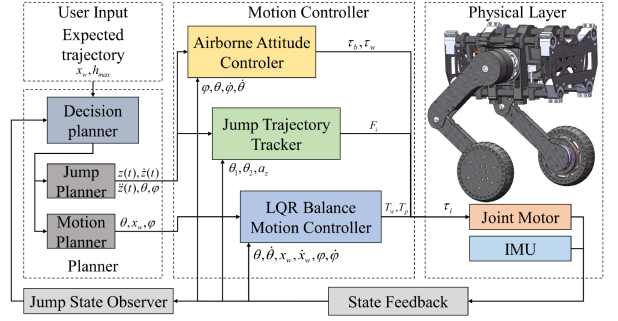


Fig. 7. Layered control framework including user input, planning, motion control, physical layer and state feedback.

the response speed, we eliminate the integral term and use a PD controller to adjust the robot's posture:

$$\begin{cases} \tau_b = k_{pb}(\varphi_0 - \varphi) + k_{db}(0 - \dot{\varphi}) \\ \tau_w = k_{pw}(\theta_0 - \theta) + k_{dw}(0 - \dot{\theta}) \end{cases} \quad (25)$$

where φ_0 and θ_0 are the body pitch angle and pendulum swing angle of the robot in the steady state.

IV. SIMULATION AND EXPERIMENT

Fig. 7 shows the control block diagram of the entire robot, with a control frequency of 500 Hz in both the simulation and prototype platforms. It is important to note that after the robot enters the airborne phase, the LQR balance controller needs to be disconnected promptly. Upon detecting ground contact, the airborne attitude controller should be disconnected in a timely manner.

A. Parameter Selection and Justification

To enhance transparency and reproducibility, the selection of key parameters and their rationale are provided as follows: The Q matrix is a state weight matrix, and the larger the diagonal elements, the higher the weight of the corresponding state variables in the cost function. This article places more emphasis on body posture control and sets $Q = \text{diag}[1000, 100, 1200, 120, 6000, 40]$. The R matrix is the control weight matrix, and the larger the diagonal elements, the stronger the constraint on the control input. In this paper, $R = \text{diag}[5.5, 5.5]$.

The setting of thresholds f_{nb} and a_{gb} requires ground experiments and verification through multiple simulations and experiments. When the robot leaves the ground, the support force and Z-axis acceleration will rapidly decay, and when it touches the ground, they will surge and rapidly decrease to a stable value. In addition, a certain margin needs to be set for the judgment of ground departure/contact to avoid misjudgment. The ground contact threshold in this article is $f_{nb} = 15\text{ N}$, $a_{gb} = 0.2\text{ g}$, and the ground departure threshold is $f_{nb} = 10\text{ N}$, $a_{gb} = -0.2\text{ g}$.

For the leg impedance controller gain settings, during the time interval from t_0 to t_2 , the gain should be set relatively high ($D_d = 50$, $K_d = 1200$) to achieve better trajectory tracking. In the t_2 to t_3 interval, to minimize the rigidity of the leg support

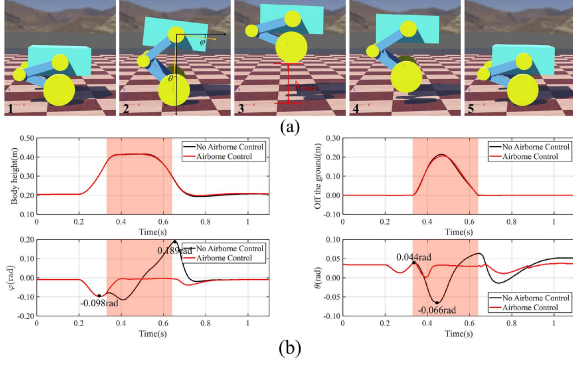


Fig. 8. Simulation of a robot jumping 20 cm in place. (a) The process of jumping 20 cm height after the robot added the airborne attitude control. (b) Comparison of robot data with and without airborne attitude control, with the aerial phase shown in the red shaded portion of the figure.

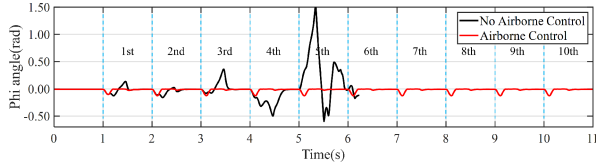


Fig. 9. Body pitch angle curve of the robot jumping in place for 10 times consecutively.

at the moment of landing and reduce impact, the gain should be set lower ($D_d = 20$, $K_d = 600$).

The remaining gains are set in this paper as: $K_p = 300$, $K_d = 30$, $K_{pz0} = 350$, $K_{dz0} = 80$, $K_{ps} = 500$, $K_{ds} = 300$, $k_{pb} = 50$, $k_{db} = 1.5$, $k_{pw} = 20$, $k_{pw} = 1$.

B. Simulation

The proposed planning and control scheme was validated in the open-source Webots simulator. The physical parameters are consistent with the actual machine.

As shown in Fig. 8(b), the aerial posture control has minimal impact on the overall jump trajectory but significantly alters the body pitch angle (ϕ) and leg swing angle (θ). Without control, the peak pitch angle reaches 0.189 rd, which is reduced to 0.098 rd (a 48.1% decrease) with control. Similarly, the peak leg swing angle drops from 0.066 rd to 0.044 rd, a 33.3% reduction. This improvement stems from the fact that, during the aerial phase, the robot's angular momentum remains conserved, and variations in leg length induce body rotation. The posture controller effectively stabilizes the orientation mid-air. Moreover, upon landing, the controlled robot exhibits significantly smaller changes in both pitch and leg angles, enhancing landing stability.

To further verify the effectiveness of the aerial posture control strategy, the robot performed 10 consecutive in-place jumps of 20 cm height at 1 s intervals. As shown in Fig. 9, without control, pitch angle fluctuations grew noticeably from the 3rd jump onward, exceeding 1.5 rd by the 5th jump and ultimately causing a fall. In contrast, with aerial posture control, the robot maintained stable orientation throughout all jumps and completed the sequence successfully. These results confirm that the

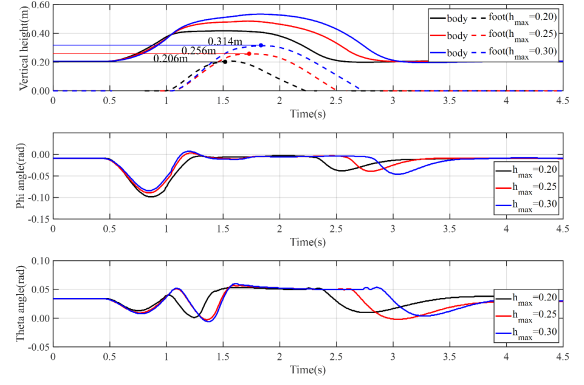


Fig. 10. Vertical position change curves of the body and foot end at different desired jump heights and change curves of the body pitch angle and leg swing angle.

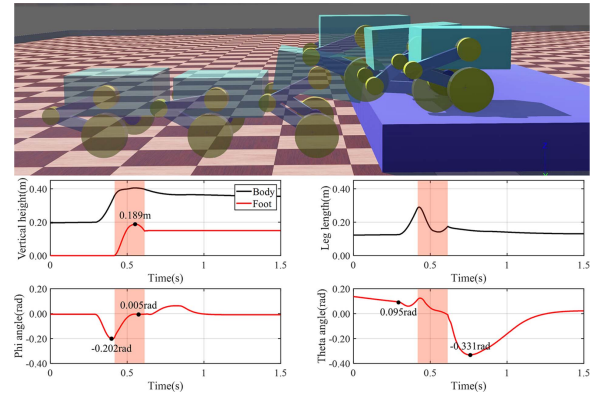


Fig. 11. The process of jumping 15 cm steps and detailed data chart.

proposed strategy significantly improves jumping stability and success rate.

To further validate the effectiveness of the jump height control, Fig. 10 presents relevant data for target heights of 0.20 m, 0.25 m, and 0.30 m. The corresponding height errors are 3% , 3.2% , and 4.67% , with the body pitch angle peak value remaining within 0.1 rd, and the leg swing angle peak value within 0.06 rd. The experiments demonstrate the accuracy of the jump height and the effectiveness of the robot's posture control.

Fig. 11 illustrates the process and data of the robot jumping onto a step. The height of the step is 15 cm, while the robot's target jump height is 20 cm, with a forward speed of 1 m/s. From the data, it can be seen that the leg swing angle is not in a balanced position at the start of the jump. However, during the aerial phase, the robot is able to quickly adjust its posture, maintaining stability and successfully completing the entire jump. This experiment further validates the reliability and robustness of the proposed jump control and airborne posture control.

C. Prototype Experiments

The robot in this study is a six-degree-of-freedom bipedal wheel-leg robot, as shown in Fig. 12, with a total mass of 9.64 kg. The core controller is based on the STM32H7, which is responsible for all motion control decisions, sensor data processing,

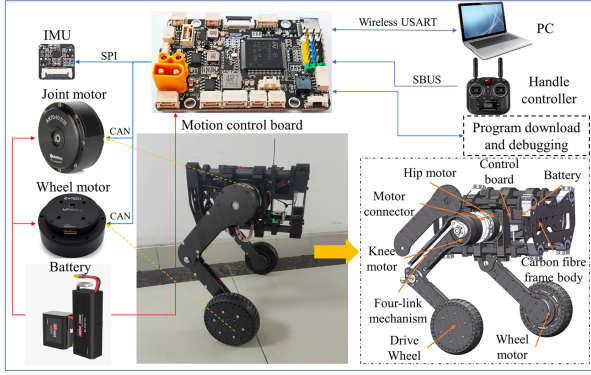


Fig. 12. Block diagram of the robot hardware system.

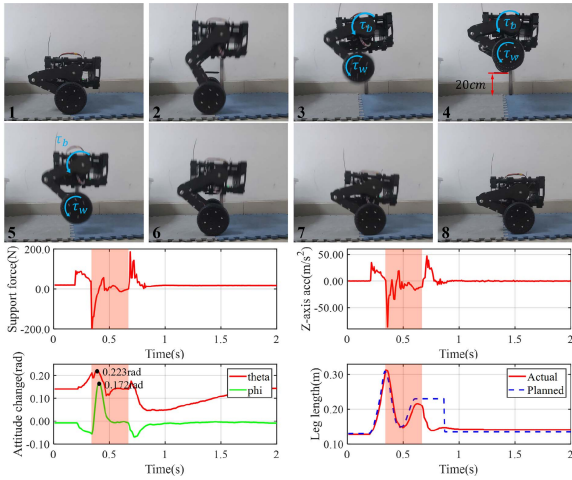


Fig. 13. Prototype in situ jump 20 cm height state response results.

and issuing control commands. Motion instructions are received via a remote controller, and robot status information is acquired through the upper-level computer. The robot is powered by two 6 s lithium-ion batteries. A six-axis Inertial Measurement Unit (IMU) is integrated into the robot's body for real-time state estimation with an acquisition frequency of 1 KHz. All joints are driven by integrated joint motors, with the hip and knee motors equipped with reducers to provide higher torque. The wheel joint motors use outer-rotor brushless motors to achieve better dynamic response and higher rotational speed. The hip and knee motors have a peak torque of 25 Nm and the wheel joint motors have a peak torque of 3 Nm.

In terms of structure, all joint axes of the robot are aligned in parallel, with a mirror-symmetric design for the two legs. The robot consists of hip joints, knee joints, and drive wheel joints. The legs adopt a serial configuration, with the knee and hip joints arranged in series and mounted within the body. A four-bar linkage mechanism is used to transmit the torque from the knee motor to the knee joint.

Fig. 13 shows the experiment of the robot jumping in place, with an expected height of 20 cm. The jumping process is consistent with the plan in Section III-A, verifying the feasibility of the plan and the effectiveness of the jumping state observer.

TABLE I
JUMP PARAMETER COMPARISON

Parameter	Proposed Robot	Compared Robot
Jump height (m)	0.20	0.10
Max pitch fluctuation (rad)	0.24	0.45
Max leg length tracking error (m)	0.02	0.03

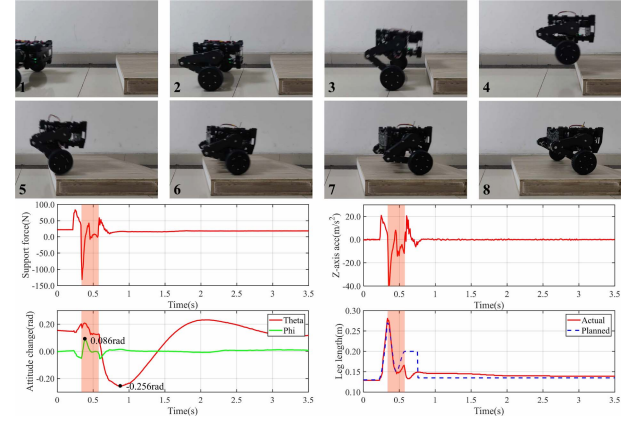


Fig. 14. The process of forward jumping 10 cm step of the prototype and the detailed data graph.

Before 0.5 seconds, the actual leg length curve matched the planned curve height, with a jump height of 20 cm, indicating good trajectory tracking performance of the leg impedance controller. After 0.5 seconds, there was a delay between the actual and planned results, mainly to reduce the rigidity of the leg support at the moment of contact with the ground and minimize the impact. Therefore, the leg length control gain was reduced during the airborne descent phase. At the moment of take-off, due to the inertia of the legs and driving wheels, there were significant fluctuations in the pitch angle and leg swing angle of the aircraft. However, after the intervention of the take-off attitude control, the pitch angle decreased rapidly from 0.172 rd to around 0 rd, and the leg swing angle decreased from 0.223 rd to around 0.140 rd, further verifying the effectiveness of the takeoff attitude control during the flight phase.

To validate the superiority of the proposed controller, we compare it with the jumping performance reported in [15], as shown in Table I. Since the robot reaches its apex around 0.5 s, and leg length predominantly influences jump height during the 0–0.5 s interval, the comparison focuses on this phase. Even when achieving twice the jump height of the robot in [15], our method reduces pitch angle variation by 46.7% and lowers the maximum leg length tracking error by 33.3%. As pitch stability is critical for robust jumping and leg tracking accuracy determines jump height, these results highlight the clear advantages of the proposed controller.

As in Fig. 14, the experiment of robot jumping steps was carried out. The height of the step is 10 cm, with a target jump height of 15 cm and a forward speed of 1 m/s. From the attitude adjustment data, it can be seen that after entering the aerial phase, the controller rapidly adjusts the robot's posture to a balanced state to counteract the impact. Upon landing, the

TABLE II
JUMP PARAMETER COMPARISON

Parameter	Proposed Robot	Compared Robot
Jump height (m)	0.15	0.15
Max pitch fluctuation (rad)	0.14	0.60
Max leg length tracking error (m)	0.01	0.05

balance controller causes the legs to swing forward to eliminate the forward inertia and quickly return to a stable state. Throughout the process, the peak value of the body pitch angle is only 0.086 rd, and the peak value of the leg swing angle is 0.256 rd. The robot successfully completes the task while maintaining excellent stability.

Similarly, a comparative analysis was conducted for the step-jumping task, as shown in Table II. The results show that the proposed robot achieved a 76.7% reduction in body pitch angle fluctuation and a 80% reduction in maximum leg length tracking error compared to the robot in [15]. These results further validate the superior performance of the proposed control strategy.

V. CONCLUSIONS AND FUTURE WORK

This paper investigates the control of jumping in a bipedal wheel-legged robot, proposing a framework that considers the realization of different jumping heights and the enhancement of stability. First, a jump trajectory planning method and an impedance-based trajectory tracking controller are developed and integrated with a jump state observer to enable high-accuracy jump control. Second, to address attitude instability during the airborne phase, an airborne dynamics model is proposed for the bipedal wheel-legged robot, and an airborne attitude control strategy is designed, which significantly improves the stability and reliability of the jumping process. Both simulation and prototype experiments validate the superior performance of the proposed approach in terms of jump accuracy and attitude stability.

In the future, we plan to add dampers to the mechanical structure to reduce the impact on the robot upon landing. Additionally, we aim to upgrade the hardware platform by incorporating sensors to enable more autonomous movement of the robot.

REFERENCES

- [1] L. Cui et al., "Learning-based balance control of wheel-legged robots," *IEEE Robot. Automat. Lett.*, vol. 6, no. 4, pp. 7667–7674, Oct. 2021.
- [2] V. Klemm et al., "Ascento: A two-wheeled jumping robot," in *Proc. IEEE Int. Conf. Robot. Automat.*, 2019, pp. 7515–7521.
- [3] V. Klemm et al., "LQR-assisted whole-body control of a wheeled bipedal robot with kinematic loops," *IEEE Robot. Automat. Lett.*, vol. 5, no. 2, pp. 3745–3752, Apr. 2020.
- [4] Ascento, "This is Ascento Pro," 2021, Accessed: Feb. 20, 2025. [Online]. Available: <https://www.youtube.com/watch?v=Uxt2wTI0m5o>
- [5] H. Zhou, H. Yu, X. Li, H. Feng, S. Zhang, and Y. Fu, "Configuration transformation of the wheel-legged robot using inverse dynamics control," in *Proc. IEEE Int. Conf. Robot. Automat.*, 2021, pp. 3091–3096.
- [6] T. Liu, C. Zhang, J. Wang, S. Song, and M. Q.-H. Meng, "Towards terrain adaptability: In situ transformation of wheel-biped robots," *IEEE Robot. Automat. Lett.*, vol. 7, no. 2, pp. 3819–3826, Apr. 2022.
- [7] D. Liu, F. Yang, X. Liao, and X. Lyu, "DIABLO: A 6-DoF wheeled bipedal robot composed entirely of direct-drive joints," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, 2024, pp. 3605–3612.
- [8] S. Wang et al., "Balance control of a novel wheel-legged robot: Design and experiments," in *Proc. IEEE Int. Conf. Robot. Automat.*, 2021, pp. 6782–6788.
- [9] H. Chen, B. Wang, Z. Hong, C. Shen, P. M. Wensing, and W. Zhang, "Underactuated motion planning and control for jumping with wheeled-bipedal robots," *IEEE Robot. Automat. Lett.*, vol. 6, no. 2, pp. 747–754, Apr. 2021.
- [10] X. Liu et al., "A novel adaptive dynamic optimal balance control method for wheel-legged robot," *Appl. Math. Modelling*, vol. 137, 2025, Art. no. 115737.
- [11] F. Raza and M. Hayashibe, "Towards robust wheel-legged biped robot system: Combining feedforward and feedback control," in *Proc. IEEE/SICE Int. Symp. System Integration*, 2021, pp. 606–612.
- [12] Y. Xin, X. Rong, Y. Li, B. Li, and H. Chai, "Movements and balance control of a wheel-leg robot based on uncertainty and disturbance estimation method," *IEEE Access*, vol. 7, pp. 133265–133273, 2019.
- [13] J. Yang, Q. Wu, S. Li, Y. Ye, and C. Luo, "Integrated modeling and control optimization of biped wheel-legged robot," *IEEE Robot. Automat. Lett.*, vol. 10, no. 2, pp. 1465–1472, Feb. 2025.
- [14] J. Yu, Z. Zhu, J. Lu, S. Yin, and Y. Zhang, "Modeling and MPC-based pose tracking for wheeled bipedal robot," *IEEE Robot. Automat. Lett.*, vol. 8, no. 12, pp. 7881–7888, Dec. 2023.
- [15] Z. Lu, Y. Xiong, H. Liu, L. Yao, and Z. Wang, "Trajectory planning for jumping and soft landing with a new wheeled bipedal robot," *IEEE Trans. Ind. Informat.*, vol. 20, no. 11, pp. 13406–13415, Nov. 2024.
- [16] Y. Zhuang et al., "Height control and optimal torque planning for jumping with wheeled-bipedal robots," in *Proc. 6th IEEE Int. Conf. Adv. Robot. Mechatron.*, 2021, pp. 477–482.
- [17] C. Yu, X. Zhu, X. Ma, K. Mao, H. Lai, and A. Yang, "Motion control study of a two wheel-leg robot with terrain adaptation capability," in *Proc. IEEE 8th Int. Conf. Vis., Image Signal Process.*, 2024, pp. 1–6.
- [18] Y. Wang, Y. Ding, and Y. Wang, "Motion control of a novel bipedal wheeled robot: VC-1," in *Proc. IEEE Int. Conf. Mechatron. Automat.*, 2024, pp. 1813–1818.
- [19] Z. Xu, J. Chen, B. Chen, P. Sun, G. Xiao, and X. Luo, "A wheeled bipedal robot capable of standing jumps," in *Proc. IEEE Int. Conf. Robot. Biomimetics*, 2024, pp. 1514–1519.
- [20] Y. Chen, H. Wang, and L. Zhang, "Control of wheel-legged balancing robot (in Chinese)," *Inf. Control*, vol. 52, no. 5, pp. 648–659, 2023. [Online]. Available: <https://xk.sia.cn/en/article/doi/10.13976/j.cnki.xk.2023.2533>
- [21] S. Sung and Y. Youm, "Landing motion control of articulated legged robot," in *Proc. IEEE Int. Conf. Robot. Automat.*, 2007, pp. 3230–3236.
- [22] Z. Cui, G. Zhang, X. Rong, and Y. Li, "Design and motion planning of hydraulically driven leg for maximum height jumping," *Mechatronics*, vol. 74, 2021, Art. no. 102499.
- [23] C. Ott, R. Mukherjee, and Y. Nakamura, "Unified impedance and admittance control," in *Proc. IEEE Int. Conf. Robot. Automat.*, 2010, pp. 554–561.
- [24] Z. Chen, S. Wang, J. Wang, and K. Xu, "Attitude stability control for multi-agent six wheel-legged robot," *IFAC-PapersOnLine*, vol. 53, no. 2, pp. 9636–9641, 2020.